# Computational Modeling of Radiation-Tolerant Photovoltaic Architectures: A DFT Approach

Yashwant Mishra National Institute of Technology (NIT), Silchar

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**Abstract**

This report outlines the computational framework and strategic execution of a device physics project focused on modeling radiation-tolerant photovoltaic architectures. Leveraging Density Functional Theory (DFT), the project systematically benchmarks the electronic, structural, and optical properties of elemental and compound semiconductors within a PIN photodiode stack. The core objective is to validate device operational stability against extra-terrestrial radiation degradation.

## 1 Strategic Objective

The primary mandate of this initiative is to computationally model and validate the performance stability of multi-layer PIN photodiode and solar cell architectures under high-radiation space environments. By establishing a robust ground-level baseline, the project seeks to isolate material vulnerabilities and engineer a device topology capable of maintaining structural integrity and power conversion efficiency in aerospace deployments.

## 2 Methodology & Material Matrix

The simulation utilizes advanced Density Functional Theory (DFT) software to execute quantum mechanical modeling. The material matrix encompasses four distinct semiconductor classes evaluated for the active absorber layer:

- **Elemental Semiconductors:** Silicon (Si), Germanium (Ge)
- **Compound Semiconductors:** Gallium Arsenide (GaAs), Gallium Phosphide (GaP)

## 3 Software Stack Architecture

Execution of this computational modeling pipeline requires a bifurcated software stack: a quantum mechanical modeling layer for material properties and a finite-element device simulation layer for macroscopic performance metrics.

### 3.1 Quantum Mechanical (DFT) Modeling Layer

- **Vienna Ab initio Simulation Package (VASP):** Commercial-grade software utilized for electronic structure calculations and quantum-mechanical molecular dynamics.
- **Quantum ESPRESSO:** Open-source suite for nanoscale electronic-structure calculations using DFT.
- **CP2K:** Solid-state physics package utilizing mixed Gaussian and plane wave (GPW) methods.

### 3.2 Device Physics Simulation Layer

- **SCAPS-1D:** One-dimensional solar cell simulator for multi-layer structures.
- **Silvaco TCAD:** 2D/3D device simulator solving Poisson, diffusion, and transport equations.

## 4 Theoretical Framework

### 4.1 Density Functional Theory (DFT)

DFT approximates the many-body Schrödinger equation by expressing ground-state properties as a functional of electron density $n(\vec{r})$:

$$\left(-\frac{\hbar^2}{2m}\nabla^2 + V_{eff}(\vec{r})\right)\psi_i(\vec{r}) = \varepsilon_i\psi_i(\vec{r}) \tag{1}$$

### 4.2 Semiconductor Device Equations

**Poisson's Equation:**

$$\nabla \cdot (\epsilon\nabla\psi) = -q(p - n + N_D^+ - N_A^-) \tag{2}$$

**Drift-Diffusion Equations:**

$$\vec{J}_n = -q\mu_n n\nabla\psi + qD_n\nabla n \tag{3}$$

$$\vec{J}_p = -q\mu_p p\nabla\psi - qD_p\nabla p \tag{4}$$

**Continuity Equations:**

$$\frac{\partial n}{\partial t} = \frac{1}{q}\nabla \cdot \vec{J}_n + G - R \tag{5}$$

$$\frac{\partial p}{\partial t} = -\frac{1}{q}\nabla \cdot \vec{J}_p + G - R \tag{6}$$

## 5 Device Topology

Top Contact $\rightarrow$ HTL $\rightarrow$ Active Layer $\rightarrow$ ETL $\rightarrow$ FTO $\rightarrow$ Bottom Contact

## 6 Environmental Stress Testing & Metrics

- I-V Characteristics
- Open-Circuit Voltage ($V_{oc}$)
- Short-Circuit Current Density ($J_{sc}$)
- Fill Factor (FF)

## 7 Quantitative Impact

- Modeled DFT profiles across 4 semiconductor classes
- Architected 5-layer simulation stack
- Validated $<$5
- Optimized material selection for aerospace use